

2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSC-901

(Mathematical Foundation of Computer Science)

Full Marks : 70

*The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.***Students can use Z table or T table as per need, provided with the question paper.**Answer **question nos. 1, 2** and **any four** from the rest.1. Answer **any five** questions :

2×5

(a) Roll two dice and consider the following events :

A = 'first die is 3', B = 'sum is 6', C = 'sum is 7'

A is independent of (i) B and C, (ii) B alone, (iii) C alone, (iv) Neither B nor C. (Justify your answer)

(b) Let A and B be two events. Suppose the probability that neither A nor B occurs is $\frac{2}{3}$. What is the probability that one or both occur?(c) Suppose that X takes values between 0 and 1 and has a probability density function is $2x$. Compute $\text{Var}(X)$ and $\text{Var}(X^2)$.

(d) You are lecturing to a group of 1000 students. You ask them to each randomly pick an integer between 1 and 10. Assuming their picks are truly random. What percentage of the students would you expect to pick a number less than or equal to 6?

(e) A teacher recorded the following test scores for her class :

[85, 78, 92, 88, 84, 90, 79, 81, 87, 93, 78, 84].

(i) Determine the median score, (ii) Identify the mode(s) if any.

(f) Suppose that the cdf of X is given by :

$$F(a) = \begin{cases} 0 & \text{for } a < 0 \\ \frac{1}{5} & \text{for } 0 \leq a < 2 \\ \frac{2}{5} & \text{for } 2 \leq a < 4 \\ 1 & \text{for } a \geq 4. \end{cases}$$

Determine the pmf of X.

(g) Find covariance for following data $x = \{6, 5, 3, 4, 2\}$, $y = \{12, 10, 8, 6, 4\}$.

Please Turn Over

(2686)

2. Answer **any five** questions :

4×5

- (a) Game : 5 red and 2 green balls in an urn. A random ball is selected and replaced by a ball of the other color; then a second ball is drawn.
- (i) What is the probability that the second ball is red?
 - (ii) What is the probability that the first ball was red, given that the second ball was red?
- (b) A multiple-choice exam has 4 choices for each question. A student has studied enough so that the probability they will know the answer to a question is 0.5. The probability that they will be able to eliminate one choice is 0.25; otherwise, all four choices seem equally plausible. If they know the answer, they will get the question right. If not, they have to guess from the three or four choices. As the teacher, you want the test to measure what the student knows. If the student answers a question correctly, what's the probability they knew the answer?

- (c) Diagonalize the following matrix, if possible $A = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 2 & 2 \\ 0 & 0 & 4 \end{bmatrix}$.

- (d) Find the Inverse of A [in question 2(c) above] using Cayley–Hamilton Theorem.
- (e) Two fair coins are flipped. As a result of this, tail and head runs occurred, where a tail run is a consecutive occurrence of at least one tail. Determine the probability function of the number of tail runs.
- (f) Suppose 100 people all toss a hat into a box and then proceed to randomly pick one out of the hats. What is the expected number of people to get their own hat back?
- (g) Explain Correlation and Covariance.

3. We draw two random cards without replacement, that is, after we draw the first card, we do not replace it in the deck. Find the probability of the following events : 2×5

- (a) Both cards are red.
- (b) At least one card is black.
- (c) The first card is black and the second red.
- (d) There is one red and one black card.
- (e) Both cards are aces.

4. (a) The Randomizer holds the 6-sided die in one fist and the 8-sided die in the other. The Roller selects one of the Randomizer's fists and covertly takes the die. The Roller rolls the die in secret and reports the result to the table. Given the reported number, what is the probability that the 6-sided die was chosen?
- (b) A fair coin is tossed repeatedly and independently. Find the expected number of tosses till the pattern HTH appears. 7+3

For a particular shop, the daily demand of an item with associated probabilities is given below

Daily demand	0	10	20	30	40	50
Probability	0.01	0.20	0.15	0.50	0.12	0.02

5. (a) The sequence of random numbers (out of 100 random numbers generated between 00 and 99) is used as 40, 19, 87, 83, 73, 84, 29, 09, 02, 20. What will be the average daily demand?
- (b) In the population, the average IQ is 100 with a standard deviation of 15. A team of scientists aims to test a new medication to determine whether it has a positive, negative or neutral effect on intelligence. A sample of 30 participants who have taken the medication has a mean of 140. Did the medication affect intelligence?

4+6

6. Assume that a man's profession can be classified as professional, skilled labourer or unskilled labourer. Assume that, of the sons of professional men, 80 percent are professional, 10 percent are skilled labourers and 10 percent are unskilled labourers. In the case of sons of skilled labourers, 60 per cent are skilled labourers, 20 percent are professionals, and 20 percent are unskilled. Finally, in the case of unskilled labourers, 50 percent of the sons are unskilled labourers and 25 percent each are in the other two categories. Assume that every man has at least one son, and form a Markov chain by following the profession of a randomly chosen son of a given family through several generations. Set up the matrix of transition probabilities. Find the probability that a randomly chosen grandson of an unskilled labourer is a professional man.

10

7. (a) Discuss the Cayley–Hamilton Theorem in the context of matrix A :

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

- (b) A personal computer has the length of time between charges of the battery, which is normally distributed with a mean of 66 hours and a standard deviation of 20 hours. What is the probability that the length of time will be between 58 and 75 hours?

5+5

t Table

cum. prob	$t_{.50}$	$t_{.75}$	$t_{.80}$	$t_{.85}$	$t_{.90}$	$t_{.95}$	$t_{.975}$	$t_{.99}$	$t_{.995}$	$t_{.999}$	$t_{.9995}$
one-tail	0.50	0.25	0.20	0.15	0.10	0.05	0.025	0.01	0.005	0.001	0.0005
two-tails	1.00	0.50	0.40	0.30	0.20	0.10	0.05	0.02	0.01	0.002	0.001
df											
1	0.000	1.000	1.376	1.963	3.078	6.314	12.71	31.82	63.66	318.31	636.62
2	0.000	0.816	1.061	1.386	1.886	2.920	4.303	6.965	9.925	22.327	31.599
3	0.000	0.765	0.978	1.250	1.638	2.353	3.182	4.541	5.841	10.215	12.924
4	0.000	0.741	0.941	1.190	1.533	2.132	2.776	3.747	4.604	7.173	8.610
5	0.000	0.727	0.920	1.156	1.476	2.015	2.571	3.365	4.032	5.893	6.869
6	0.000	0.718	0.906	1.134	1.440	1.943	2.447	3.143	3.707	5.208	5.959
7	0.000	0.711	0.896	1.119	1.415	1.895	2.365	2.998	3.499	4.785	5.408
8	0.000	0.706	0.889	1.108	1.397	1.860	2.306	2.896	3.355	4.501	5.041
9	0.000	0.703	0.883	1.100	1.383	1.833	2.262	2.821	3.250	4.297	4.781
10	0.000	0.700	0.879	1.093	1.372	1.812	2.228	2.764	3.169	4.144	4.587
11	0.000	0.697	0.876	1.088	1.363	1.796	2.201	2.718	3.106	4.025	4.437
12	0.000	0.695	0.873	1.083	1.356	1.782	2.179	2.681	3.055	3.930	4.318
13	0.000	0.694	0.870	1.079	1.350	1.771	2.160	2.650	3.012	3.852	4.221
14	0.000	0.692	0.868	1.076	1.345	1.761	2.145	2.624	2.977	3.787	4.140
15	0.000	0.691	0.866	1.074	1.341	1.753	2.131	2.602	2.947	3.733	4.073
16	0.000	0.690	0.865	1.071	1.337	1.746	2.120	2.583	2.921	3.686	4.015
17	0.000	0.689	0.863	1.069	1.333	1.740	2.110	2.567	2.898	3.646	3.965
18	0.000	0.688	0.862	1.067	1.330	1.734	2.101	2.552	2.878	3.610	3.922
19	0.000	0.688	0.861	1.066	1.328	1.729	2.093	2.539	2.861	3.579	3.883
20	0.000	0.687	0.860	1.064	1.325	1.725	2.086	2.528	2.845	3.552	3.850
21	0.000	0.686	0.859	1.063	1.323	1.721	2.080	2.518	2.831	3.527	3.819
22	0.000	0.686	0.858	1.061	1.321	1.717	2.074	2.508	2.819	3.505	3.792
23	0.000	0.685	0.858	1.060	1.319	1.714	2.069	2.500	2.807	3.485	3.768
24	0.000	0.685	0.857	1.059	1.318	1.711	2.064	2.492	2.797	3.467	3.745
25	0.000	0.684	0.856	1.058	1.316	1.708	2.060	2.485	2.787	3.450	3.725
26	0.000	0.684	0.856	1.058	1.315	1.706	2.056	2.479	2.779	3.435	3.707
27	0.000	0.684	0.855	1.057	1.314	1.703	2.052	2.473	2.771	3.421	3.690
28	0.000	0.683	0.855	1.056	1.313	1.701	2.048	2.467	2.763	3.408	3.674
29	0.000	0.683	0.854	1.055	1.311	1.699	2.045	2.462	2.756	3.396	3.659
30	0.000	0.683	0.854	1.055	1.310	1.697	2.042	2.457	2.750	3.385	3.646
40	0.000	0.681	0.851	1.050	1.303	1.684	2.021	2.423	2.704	3.307	3.551
60	0.000	0.679	0.848	1.045	1.296	1.671	2.000	2.390	2.660	3.232	3.460
80	0.000	0.678	0.846	1.043	1.292	1.664	1.990	2.374	2.639	3.195	3.416
100	0.000	0.677	0.845	1.042	1.290	1.660	1.984	2.364	2.626	3.174	3.390
1000	0.000	0.675	0.842	1.037	1.282	1.646	1.962	2.330	2.581	3.098	3.300
Z	0.000	0.674	0.842	1.036	1.282	1.645	1.960	2.326	2.576	3.090	3.291
	0%	50%	60%	70%	80%	90%	95%	98%	99%	99.8%	99.9%
	Confidence Level										

STANDARD NORMAL DISTRIBUTION: Table Values Represent AREA to the LEFT of the Z score.

Z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.9	.00005	.00005	.00004	.00004	.00004	.00004	.00004	.00004	.00003	.00003
-3.8	.00007	.00007	.00007	.00006	.00006	.00006	.00006	.00005	.00005	.00005
-3.7	.00011	.00010	.00010	.00010	.00009	.00009	.00008	.00008	.00008	.00008
-3.6	.00016	.00015	.00015	.00014	.00014	.00013	.00013	.00012	.00012	.00011
-3.5	.00023	.00022	.00022	.00021	.00020	.00019	.00019	.00018	.00017	.00017
-3.4	.00034	.00032	.00031	.00030	.00029	.00028	.00027	.00026	.00025	.00024
-3.3	.00048	.00047	.00045	.00043	.00042	.00040	.00039	.00038	.00036	.00035
-3.2	.00069	.00066	.00064	.00062	.00060	.00058	.00056	.00054	.00052	.00050
-3.1	.00097	.00094	.00090	.00087	.00084	.00082	.00079	.00076	.00074	.00071
-3.0	.00135	.00131	.00126	.00122	.00118	.00114	.00111	.00107	.00104	.00100
-2.9	.00187	.00181	.00175	.00169	.00164	.00159	.00154	.00149	.00144	.00139
-2.8	.00256	.00248	.00240	.00233	.00226	.00219	.00212	.00205	.00199	.00193
-2.7	.00347	.00336	.00326	.00317	.00307	.00298	.00289	.00280	.00272	.00264
-2.6	.00466	.00453	.00440	.00427	.00415	.00402	.00391	.00379	.00368	.00357
-2.5	.00621	.00604	.00587	.00570	.00554	.00539	.00523	.00508	.00494	.00480
-2.4	.00820	.00798	.00776	.00755	.00734	.00714	.00695	.00676	.00657	.00639
-2.3	.01072	.01044	.01017	.00990	.00964	.00939	.00914	.00889	.00866	.00842
-2.2	.01390	.01355	.01321	.01287	.01255	.01222	.01191	.01160	.01130	.01101
-2.1	.01786	.01743	.01700	.01659	.01618	.01578	.01539	.01500	.01463	.01426
-2.0	.02275	.02222	.02169	.02118	.02068	.02018	.01970	.01923	.01876	.01831
-1.9	.02872	.02807	.02743	.02680	.02619	.02559	.02500	.02442	.02385	.02330
-1.8	.03593	.03515	.03438	.03362	.03288	.03216	.03144	.03074	.03005	.02938
-1.7	.04457	.04363	.04272	.04182	.04093	.04006	.03920	.03836	.03754	.03673
-1.6	.05480	.05370	.05262	.05155	.05050	.04947	.04846	.04746	.04648	.04551
-1.5	.06681	.06552	.06426	.06301	.06178	.06057	.05938	.05821	.05705	.05592
-1.4	.08076	.07927	.07780	.07636	.07493	.07353	.07215	.07078	.06944	.06811
-1.3	.09680	.09510	.09342	.09176	.09012	.08851	.08691	.08534	.08379	.08226
-1.2	.11507	.11314	.11123	.10935	.10749	.10565	.10383	.10204	.10027	.09853
-1.1	.13567	.13350	.13136	.12924	.12714	.12507	.12302	.12100	.11900	.11702
-1.0	.15866	.15625	.15386	.15151	.14917	.14686	.14457	.14231	.14007	.13786
-0.9	.18406	.18141	.17879	.17619	.17361	.17106	.16853	.16602	.16354	.16109
-0.8	.21186	.20897	.20611	.20327	.20045	.19766	.19489	.19215	.18943	.18673
-0.7	.24196	.23885	.23576	.23270	.22965	.22663	.22363	.22065	.21770	.21476
-0.6	.27425	.27093	.26763	.26435	.26109	.25785	.25463	.25143	.24825	.24510
-0.5	.30854	.30503	.30153	.29806	.29460	.29116	.28774	.28434	.28096	.27760
-0.4	.34458	.34090	.33724	.33360	.32997	.32636	.32276	.31918	.31561	.31207
-0.3	.38209	.37828	.37448	.37070	.36693	.36317	.35942	.35569	.35197	.34827
-0.2	.42074	.41683	.41294	.40905	.40517	.40129	.39743	.39358	.38974	.38591
-0.1	.46017	.45620	.45224	.44828	.44433	.44038	.43644	.43251	.42858	.42465
-0.0	.50000	.49601	.49202	.48803	.48405	.48006	.47608	.47210	.46812	.46414

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COMPUTER SCIENCE AND ENGINEERING

Paper : CSC-902

(Advanced Algorithms)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question nos. 1, 2** and **any four** questions from the rest.

1. Answer **any five** questions : 2×5
 - (a) Compute the prefix function used in the KMP algorithm for the pattern string 'cgtacgttcgtac'.
 - (b) Draw a standard trie for the following set of strings :
 $\{abab, baba, ccccc, bbaaaa, caa, bbaacc, cbcc, cbca\}$.
 - (c) State the differences between k-D tree and range tree in orthogonal searching.
 - (d) Explain why the randomized closest pair algorithm uses hashing to store the points.
 - (e) Give the polynomial time reduction algorithm for $SAT \leq_p Clique$.
 - (f) When would you use the trie data structure and when the KMP algorithm for pattern matching?
 - (g) Define NP-hard and NP-complete problems.
2. Answer **any five** questions : 4×5
 - (a) State and prove Max-flow Min-cut Theorem.
 - (b) What will happen if in Divide and Conquer Select Algorithm to find k th smallest element, the true median of the elements is chosen as the pivot instead of the median of medians? Explain.
 - (c) Let $x = a + bi$ and $y = c + di$ be two complex numbers. The product xy can easily be calculated using four multiplications, that is, $xy = (ac - bd) + (ad + bc)i$. Devise a method for computing the product xy using only three multiplications.
 - (d) Prove that the Merging step of Divide and Conquer based closest pair algorithm runs in $O(n)$ time.
 - (e) Give the dynamic programming formulation of optimal binary search tree generation problem with explanation.
 - (f) In Divide and Conquer Select algorithm to find k th smallest element, if the group size is g , derive the time complexity of the algorithm in terms of g .
 - (g) Give a dynamic programming formulation of the following problem : Given a grid of size $m \times n$. Count the total number of unique paths from the top-left to the bottom-right corner of the grid.

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3. (a) Given an array $A[1..n]$ of n elements, where n is a power of 2, give an algorithm to find both the minimum and maximum of the elements such that it does exactly $2n - 3$ comparisons. Justify your answer.
- (b) Give a Divide and Conquer algorithm to find both minimum and maximum of the elements and show that it takes exactly $(3n/2) - 2$ comparisons. 5+5
4. (a) Define NP Class.
- (b) Define travelling salesperson problem(TSP), and prove that $TSP \in NP$.
- (c) Given problems $P1$ and $P2$. Describe the significance of $P1 \leq_p P2$. 2+3+5
5. (a) Give an approximation algorithm along with data structure and time complexity to solve the Travelling Salesperson Problem, assuming triangular inequality.
- (b) Derive the approximation bound for the above algorithm. 5+5
6. (a) In a randomized closest pair algorithm, prove that the points p_j in a 5×5 grid centering the grid of the current point p_i is sufficient to identify the closest pair till i points are processed.
- (b) Derive the average case expected time complexity of randomized quicksort algorithm. 5+5
7. (a) Given a flow network $G = (V, E)$, $V = \{s, t, p, q, x, y\}$ the set of vertices, and $E = \{(s, p, 5), (s, q, 3), (p, t, 3), (p, y, 3), (p, q, 4), (q, x, 4), (q, y, 1), (x, y, 3), (x, t, 2), (y, t, 3)\}$ the set of edges with capacities. Give the maximum flow in the network along with the corresponding cut.
- (b) There are m teachers in a department and n students to whom n projects are to be allocated under supervision of m teachers. The department wants the students to be equally distributed among the teachers with the constraint that the topics offered by the teachers and topics of interest given by the students match. Each teacher can give at most k subjects and students can give up to 3 different topics of interest. Each student must be assigned to one teacher. Formulate a network flow problem such that the department can maximize its goal of equal distribution of students to each teacher. 5+5
8. (a) Prove that the expected search time complexity of Skip list is $O(\log n)$ with high probability.
- (b) Give an appropriate data structure and $O(\log n + k)$ algorithm to count the number of elements that fall in a given range $[a : b]$, where k is the count. Justify the time complexity. 5+5
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2025**COMPUTER SCIENCE AND ENGINEERING****Paper : CSE-901****(Machine Learning)****Full Marks : 70***The figures in the margin indicate full marks.**Candidates are required to give their answers in their own words as far as practicable.*Answer **question numbers 1, 2** and **any four** questions from the rest.**1. Answer any five questions :****2×5**

- (a) Define type II error for hypothesis testing.
- (b) How do you select essential features/variables while working on a dataset?
- (c) What are one-hot, label and target encoding?
- (d) What is a bias-variance trade-off? What are its implications?
- (e) Differentiate between regression and classification.
- (f) What are log-scaling, normalization and standardization?
- (g) Write the equation to find the variance of the estimates of coefficients in Logistic Regression.
- (h) What do you mean by entropy and what are its uses?

2. Answer any five questions :**4×5**

- (a) Suppose the classifier's prediction is given as follows :

		Predicted	
Actual		+	-
	+	60 TP	15 FN
	-	10 FP	15 TN

TP
 $\frac{TP}{TP + FN}$

Calculate accuracy, weighted accuracy, precision and recall.

- (b) What are the different terminating conditions for decision trees? Explain your answer.
- (c) How to show or prove a dataset is not linearly separable? Briefly explain.
- (d) What are Loss Functions and Cost Functions? Explain the critical differences between them.

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- (e) You are trying to split a node of a decision tree based on an attribute A having k categorical values. Write an algorithm to perform the best split based on the Gini index. Assume there are n classes.
- (f) Compare the classification methods : logistic regression and decision trees.
- (g) What is a recommender system and how is it implemented?
- (h) How will you justify that the designed and optimized linear regression model will be suitable for other datasets?

3. (a) Build and explain the decision tree for the following dataset (up to 2 levels) :

Tid	Refund	Material Status	Taxable Income	Cheat
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Married	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

(b) Compare and contrast Gini Impurity and Information Gain.

6+4

4. (a) What is 'K' in the K-means algorithm?

(b) How do we decide the value of 'K' in the K-means algorithm?

(c) Why is the K-means algorithm not a lazy learner?

(d) Why is scaling required in the K-means algorithm?

(e) Write down the pseudocode of the K-means algorithm.

1+2+1+2+4

5. (a) What are the significances of PCA?

(b) Write pseudocode to transform a dataset by applying PCA where the number of principal components is $k < 10$.

(c) What is the justification for choosing the eigenvectors corresponding to the largest eigenvalues for making the transform?

2+4+4

6. (a) What is meant by feature extraction? Write an algorithm to extract the features from a given dataset.
(b) Assume you are given a dataset in the form of an $(d \times n)$ matrix X for n tuples, each with d features. Prove that your k-means algorithm will not run indefinitely if the terminating condition is no change of labels. 4+6
7. (a) What is Linear Regression and what is its fundamental purpose?
(b) What is Multicollinearity and why is it a problem in Multiple Linear Regression?
(c) What is the p-value in the context of a regression coefficient?
(d) How do you know if your Linear Regression model is overfit? 2+3+2+3
8. (a) What is Logistic Regression and why is it called 'Regression' if it's used for classification?
(b) Explain the role of the Sigmoid Function (or Logit Function) in Logistic Regression.
(c) Write a short note on the basic model of a neural network and its working principle. 2+2+6
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2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSE-903

(Wireless and Mobile Computing)

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer *question nos. 1 & 2* and *any four* questions from the rest.

1. Answer *any five* questions :

2×5

- (a) State the advantage of using MILD back-off time in MACA/W compared to MACA.
- (b) State the metrics to measure the performance of a MAC protocol.
- (c) State the causes for generating immediate advertisement in DSDV. Why is it not periodic?
- (d) Is there a chance for having a collision between route request packets in DSR? If yes, how it can be avoided?
- (e) How does sectoring improve frequency reuse and reduce co-channel interference?
- (f) Derive the relation between the physical length of a $\lambda/2$ dipole and the frequency of the signal it radiates.
- (g) A wireless link operates at 2 GHz over a distance of 1 km. Using the free-space path loss formula, express the path loss in dB (no need to compute).

2. Answer *any five* questions :

4×5

- (a) Describe exposed station problem. Is the MACA able to address the problem? Justify your answer.
- (b) Illustrate, with an example, the utility of using 'Sequence Number' to prevent loop in DSDV.
- (c) Suggest a routing protocol that can be used for 'search and rescue' by a disaster management group. State reason(s) in favour of your answer.
- (d) Describe the role of PCF; even if DCF exists in the same layer for medium access. Justify your answer.
- (e) Suppose a mobile node with IP 10.10.20.35/8 moves to another network and received and configured a new IP 192.35.40.23/24. Describe how the node discovers the agents, registers and perform tunnelling in this context.

Please Turn Over

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- (f) Derive the relation between the reuse distance D and cluster size N in a cellular network.
- (g) Compute the total number of TDMA frames in one GSM hyperframe. If each time slot carries 156.25 bits (gross), compute the total number of bits transmitted in one hyperframe.
3. (a) Define Directed Acyclic Graph (DAG).
- (b) How is the link reversal algorithm used in a DAG to explore multipath routes towards destination? Explain with an example.
- (c) State the advantages and disadvantages does multipath routing have. Do you suggest a network where this technique performs better than others? Give reasons. 1+6+3
4. (a) What is the motivation behind creating zone in ZRP instead of using a flat network?
- (b) What will be the situation if the destination is not within the zone?
- (c) "Sensor networks are clustered in general." – Comment on the statement with justification. 3+5+2
5. (a) Discuss the causes of energy waste in WSN and state the role of distributed duty cycle in this context.
- (b) How does the S-MAC protocol attempt to reduce collisions?
- (c) How does the SPIN family of protocols address the basic challenges faced by flooding? 3+3+4
6. A cellular network uses hexagonal cells and operates at 2 GHz. The cell radius $R=1$ km, total available spectrum = 25 MHz and cluster size $K = 7$.
- (a) Compute frequency reuse distance.
- (b) Calculate the number of channels per cell if each channel occupies 200 kHz.
- (c) Discuss how reducing cluster size affects spectral efficiency and co-channel interference.
- (d) Suggest two practical methods to reduce co-channel interference without changing the cluster size. 3+2+3+2
7. (a) Draw the GSM network architecture and briefly explain the roles of BTS, BSC, MSC, HLR and VLR.
- (b) Explain the difference between WCDMA and GSM TDMA in terms of multiple access.
- (c) Draw and explain the E-UTRAN and EPC architecture, highlighting the role of eNodeB and MME in LTE. 3+3+4
8. (a) Describe how I-TCP splits a TCP connection at the foreign agent and maintains end-to-end connections during handover. Mention one advantage and one disadvantage of this splitting.
- (b) Explain the concept of S-TCP and how it uses local retransmissions at base stations. Discuss why S-TCP maintains end-to-end TCP semantics. 5+5
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2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSA-901

(Research Methodology and IPR)

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any five** questions from the rest.

1. Answer *any twenty* questions (choose the correct answers) : **1×20**

- (a) What is research?
- (i) Searching again and again
 - (ii) Finding a solution to any problem
 - (iii) Working in a scientific way to search for the truth of any problem
 - (iv) None of the above.
- (b) Which of the following is the first step in starting the research process?
- (i) Searching sources of information to locate the problem
 - (ii) Survey of related literature
 - (iii) Identification of the problem
 - (iv) Searching for solutions to the problem.
- (c) Manipulation is always a part of
- (i) Historical research
 - (ii) Fundamental research
 - (iii) Descriptive research
 - (iv) Experimental research.
- (d) The research that explores new facts through the study of the past is called
- (i) Philosophical research
 - (ii) Historical research
 - (iii) Mythological research
 - (iv) Content analysis.
- (e) A research problem is not feasible only when
- (i) it is researchable
 - (ii) it is new and adds something to the knowledge
 - (iii) it consists of independent and dependent variables
 - (iv) it has utility and relevance.

Please Turn Over

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- (f) Which process is not needed in experimental research?
- (i) Observation
 - (ii) Manipulation and replication
 - (iii) Controlling
 - (iv) Reference collection.
- (g) The experimental study is based on
- (i) The manipulation of variables
 - (ii) Conceptual parameters
 - (iii) Replication of research
 - (iv) Survey of literature.
- (h) Formulation of hypothesis may not be required in
- (i) Survey method
 - (ii) Historical studies
 - (iii) Experimental studies
 - (iv) Normative studies.
- (i) A patent is an exclusive right granted for
- (i) Book
 - (ii) Innovation
 - (iii) Land
 - (iv) None of these.
- (j) The 'Priority Date' of a patent is the date on which
- (i) the patent is granted
 - (ii) the patent application is first filed
 - (iii) the patent application is filed with completed specification.
 - (iv) None of the above.
- (k) Fundamental research reflects the ability to
- (i) synthesize new ideals
 - (ii) unfurl new principles
 - (iii) evaluate the existing material concerning research
 - (iv) study the existing literature regarding various topics.
- (l) The main characteristic of scientific research is
- (i) empirical
 - (ii) theoretical
 - (iii) experimental
 - (iv) All of these.
- (m) A null hypothesis is
- (i) when there is no difference between the variables
 - (ii) the same as research hypothesis
 - (iii) subjective in nature
 - (iv) when there is difference between the variables.
- (n) Generalised conclusion on the basis of a sample is technically known as
- (i) Data analysis and interpretation
 - (ii) Parameter inference
 - (iii) Statistical inference
 - (iv) All of these.

- (o) Field-work-based research is called
- (i) Empirical research
 - (ii) Historical research
 - (iii) Experimental research
 - (iv) Biographical research.
- (p) Which of the following statements is correct?
- (i) Objectives of research are stated in first chapter of the thesis.
 - (ii) Researcher must possess analytical ability.
 - (iii) The title of a research paper should briefly contain all pertinent information about the research work.
 - (iv) All of the above.
- (q) What does plagiarism mean?
- (i) Following other authors' research work and write literature review.
 - (ii) Writing praiseworthy words in support of a research paper in the introduction.
 - (iii) Copying phrases from other research papers and writing verbatim in own research papers
 - (iv) All of the above.
- (r) Which of the following is/are motive(s) of research?
- (i) Excitement of discover
 - (ii) In depth knowledge on a subject
 - (iii) Need for publication
 - (iv) All of these.
- (s) What does IPR stand for?
- (i) Innovative Property Right
 - (ii) Intelligent Property Right
 - (iii) Intellectual Property Right
 - (iv) International Property Right.
- (t) Which one of the following is a benefit of research?
- (i) Expanding frontiers of knowledge
 - (ii) New inventions and discoveries
 - (iii) Infotainment
 - (iv) All of these.
- (u) Primary sources of a research problem are
- (i) Existing literature
 - (ii) Sophisticated technologies and innovations
 - (iii) Observations of everyday life
 - (iv) None of these.
2. (a) What do you understand by the word 'research'?
- (b) What are the main motives for research?
- (c) Distinguish between fundamental research and applied research. 3+4+3
3. (a) How do you formulate a research problem?
- (b) What considerations should a researcher keep in mind while formulating a research question?
- (c) Discuss the role of literature review in approaching a research problem. 3+4+3

Please Turn Over

4. Define and distinguish between Research Methods and Research Methodology with examples. 10
- ✓5. (a) What are the characteristics of a good quality research?
(b) Elaborate different sources of a research problem. 5+5
- ✓6. (a) What are the major sources of errors in selecting a research problem?
(b) Write a short note on scopes and objectives of a research problem. 5+5
- ✓7. (a) Write down the roles of intuition in research.
(b) Explain different research variables with examples. 4+6
- ✓8. (a) What do you understand by IP and IPR?
(b) Name the major types of IPs with examples.
(c) Define and exemplify what a patent is. 4+3+3
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2025

COMPUTER SCIENCE AND ENGINEERING

Paper : CSA-902

(English for Research Paper Writing)

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any five** questions from the rest.

1. Answer **any twenty** questions (choose the correct answers) : 1×20
- (a) Bibliography is printed
- (i) at the beginning of the research paper.
 - (ii) anywhere in the research paper.
 - (iii) at the end of the research paper arranged in number or by alphabetical order.
 - (iv) not necessary to be included in a research paper.
- (b) What does plagiarism mean?
- (i) Following other authors' research work and write literature review
 - (ii) Writing praiseworthy words in support of a research paper in the introduction
 - (iii) Copying phrases from other research papers and writing verbatim in own research papers
 - (iv) All of the above.
- (c) Which of the following appear at the top of a research paper?
- (i) Introduction
 - (ii) Abstract
 - (iii) Title
 - (iv) List of abbreviations.
- (d) Formatting, spaces, spellings and grammar becomes part of
- (i) analysis
 - (ii) proofreading
 - (iii) typesetting
 - (iv) None of these.
- (e) What is not taught in a course on Research Paper Writing?
- (i) Think about a topic openly and coherently
 - (ii) Critically read and review research papers
 - (iii) Learn how to write ad campaigns
 - (iv) Write concisely with clarity and without ambiguity.

Please Turn Over

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- (f) Planning and preparation for writing a research paper include
- (i) think about why you want to publish your research.
 - (ii) give yourself enough time to plan and write your manuscript.
 - (iii) choose one paper as a model and note down useful phrases.
 - (iv) All of the above.
- (g) Choosing a journal involves
- (i) going to a book shop and select one.
 - (ii) searching the Internet by the keywords of your topic.
 - (iii) searching the wikipedia and identifying your topic.
 - (iv) All of the above.
- (h) While writing a research paper, word order in English language
- (i) is not at all important.
 - (ii) matches with that of any other language.
 - (iii) is paramount for writing a good research paper.
 - (iv) None of the above.
- (i) In any writing project, the longer your sentence
- (i) the lesser the chance it will be misunderstood.
 - (ii) the greater the chance it will be misunderstood.
 - (iii) the greater the chance it will be considered a good project.
 - (iv) None of the above.
- (j) Being concise and removing redundancy
- (i) is the best policy of writing.
 - (ii) is worst policy of writing.
 - (iii) has no connection with writing.
 - (iv) All of the above.
- (k) Highlighting your research findings requires
- (i) you have a friendship with the editor.
 - (ii) using a highlighter pen.
 - (iii) making your sentences shorter than normal.
 - (iv) None of the above.

- (l) To have hedging and avoid criticisms require
- (i) toning down verbs.
 - (ii) toning down adjectives and adverbs.
 - (iii) toning down strong claims by inserting adverbs
 - (iv) All of the above.
- (m) The introduction of a research paper should state which of the following?
- (i) Nature of the study
 - (ii) Aims
 - (iii) Sources of information
 - (iv) All of these.
- (n) A good title of a research paper has the following characteristics :
- (i) It will easily understood by the reviewers/referees and the readers.
 - (ii) It is typeset using a bold type and bigger fonts.
 - (iii) It is short.
 - (iv) All of the above.
- (o) Which of the following is/are not a form of research report?
- (i) Thesis
 - (ii) Whitepaper
 - (iii) Precise
 - (iv) Dissertation.
- (p) To avoid plagiarism, one can use the following :
- (i) Paraphrasing
 - (ii) Bold fonts
 - (iii) Slanted fonts
 - (iv) Quotation marks.
- (q) Before writing a research paper, one must
- (i) learn how to typeset using a computer.
 - (ii) learn to write with a fountain pen.
 - (iii) download the style guide of the journal.
 - (iv) None of the above.
- (r) The abstract and the conclusion sections of a research paper are
- (i) identical
 - (ii) similar but not identical
 - (iii) completely different
 - (iv) None of these.
- (s) References and bibliography
- (i) are identical
 - (ii) are totally different
 - (iii) are synonymous
 - (iv) None of these.

(t) Citation and reference are

- (i) same
- (ii) different
- (iii) antonyms
- (iv) None of these.

(u) What abbreviation is used to mention more than two authors of a research work to be cited?

- (i) at al.
- (ii) et all.
- (iii) et al.
- (iv) ot all.

(v) ISSN stands for

- (i) International Standard Social Number
- (ii) International Source Serial Number
- (iii) International Standard Serial Number
- (iv) Indian Standard Society Number.

(w) Literature review is basically to bridge the gap between

- (i) newly established facts.
- (ii) previously established facts.
- (iii) facts established time to time.
- (iv) previous to current established facts.

2. (a) Write and explain the main types of abstracts in research papers.

(b) What are the characteristics of a good introduction of a research paper? 5+5

3. (a) What is a literature review and how to write a good literature review?

(b) How should you structure the research methods in a research paper? 5+5

4. (a) Explain how to write the results of research in a research paper.

(b) Elaborate the writing strategy to highlight the key results. 5+5

5. (a) Why do you need discussion section in a research paper?

(b) How to structure the discussion section of a research paper?

(c) What factors can influence the writing of the discussion section? 2+4+4

6. (a) Define a conclusion section of a research paper.

(b) What is the difference between the conclusion and the abstract of a research paper?

(c) Where and when do we need an extended abstract and summary? 2+4+4

7. (a) What is plagiarism?

(b) How to spot plagiarism in a manuscript of a research paper?

(c) How to avoid plagiarism in your research paper writing? 2+2+6

8. We reproduce a part of the abstract of a paper from the public domain in computer science below :

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of

- (a) Try to write a suitable title of the paper.
- (b) Try to rewrite the portion of the abstract by paraphrasing in your own language to avoid plagiarism.
- (c) Write a few keywords that best describe the research work. 3+5+2

A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.